Classical Mechanics

Leonard Susskind

Core track, 2011 Fall Modern Physics Stanford Partial
Transcript-derived notes curated by LazyingArt LLC with Video2Book

Original lectures by Leonard Susskind. Transcript-derived course notes curated by LazyingArt LLC with Video2Book.

Contents

1 Classical Mechanics and the Meaning of State **1**

1.1 Opening Motivation and the Stroboscopic Toy World . . . 1

1.2 Two States, One Law, and the Meaning of Phase Space . . . 1

1.2.1 Question & Answer . . . 2

1.3 Arrow Diagrams and the First Deterministic Laws . . . 2

1.4 Allowed and Forbidden Laws in Finite State Spaces . . . 4

1.4.1 Question & Answer . . . 5

1.5 Infinite State Spaces, Disconnected Components, and Conservation as Memory . . . 5

1.6 Continuous Motion and Why Position Alone Is Not Enough . . . 6

1.6.1 Question & Answer . . . 6

1.7 First-Order Versus Second-Order Equations, and the Return to Two-Step Memory . 7

1.7.1 Question & Answer . . . 8

1.8 Summary . . . 9

2 From Initial Data to Least Action **10**

2.1 First-Order and Second-Order Laws . . . 10

2.1.1 Question & Answer . . . 11

2.2 Conservative Forces and Energy Conservation . . . 11

2.3 Force as the Time Derivative of Momentum . . . 14

2.3.1 Question & Answer . . . 16

2.4 A Mathematical Warm-Up: Stationary Points . . . 16

2.4.1 Question & Answer . . . 18

2.5 Local Motion, Whole Paths, and Functionals . . . 18

2.6 The Action . . . 19

2.6.1 Question & Answer . . . 20

2.7 Summary . . . 20

3 Least Action, Euler–Lagrange, and Symmetry **22**

3.1 The Common Form of Physical Law 22

3.2 Two Mathematical Tools Before the Main Derivation 23

3.2.1 Question & Answer 23

3.3 Trajectories, Histories, and the Local–Global Tension 24

3.4 Varying the Path and Deriving the Euler–Lagrange Equation 24

3.5 What the Euler–Lagrange Equation Says 27

3.5.1 Question & Answer 27

3.6 Worked Newtonian Examples and Translation Symmetry 28

3.7 Two Coordinates, One Broken Symmetry, and Rotational Invariance 29

3.7.1 Question & Answer 31

3.8 Summary 32

Chapter 1

Classical Mechanics and the Meaning of State

Classical mechanics enters this course with an unusually broad claim. Leonard Susskind begins by saying that it is the basis of all physics, not only because it describes ordinary moving objects, but because it provides the general architecture by which physical systems evolve. These companion notes follow that opening lecture closely and preserve a small amount of board evidence, with credit to LazyingArt LLC and Video2Book, when the geometry of the board helps stabilize the mathematical story. The strategy of the lecture is deliberate: before we study realistic particles and forces, we first strip the subject to its simplest logical core and ask what a law of evolution can mean.

1.1 Opening Motivation and the Stroboscopic Toy World

The lecture does not start with Newton's law. It starts with a simplification. Time, for the moment, will not be continuous. We imagine a stroboscopic world in which we look only at equally spaced beats, say one second apart. If q_n denotes the state of the system at the n-th beat, then the law of motion takes the schematic form

$$q_n \mapsto q_{n+1}. \tag{1.1}$$

This is the first important reduction. Instead of worrying about smooth motion, we ask only: given the present state, what is the next one?

The second simplification is to consider systems with only a very small number of configurations. Such systems are less realistic than anything in nature, but that is exactly why they are useful here. They let us separate the structure of the law from the complications of the world to which the law will later be applied.

1.2 Two States, One Law, and the Meaning of Phase Space

The first nontrivial example is a coin on a table. It has two possible configurations, heads and tails. The phase space is therefore

$$\Gamma = \{H, T\}. \tag{1.2}$$

Definition 1.1. The phase space of a system is the space of its possible states.

Definition 1.2. A state is a complete specification of the information needed to determine what happens next.

At this stage the phase space is not ordinary space. It is simply the set of possibilities. The coin gives us two points in an abstract state space, one labelled H and one labelled T.

A useful limiting case helps fix the idea. If there were only one state, then the world would be completely trivial. Nothing could happen, because there would be nowhere else to go. With two states, by contrast, there is room for genuine laws of evolution.

1.2.1 Question & Answer

Question. What exactly counts as a state?

Answer. A state is not just any description we happen to write down. It is exactly enough information to determine the next step. If the symbols H and T are enough, then those symbols are the states. If they are not enough, then the true state space must be larger. This operational definition is the thread that runs through the whole lecture.

1.3 Arrow Diagrams and the First Deterministic Laws

Once the two states are on the board, the lecture introduces its visual grammar. We represent the law by an arrow leading from the present state to the next one. The arrow is not decoration; it is the law of motion itself.

The first law is the dull one:

$$H \mapsto H, \qquad T \mapsto T. \tag{1.3}$$

If the system is in the state H now, it remains in the state H at the next beat. If it is in the state T, it remains in the state T.

The second law is more interesting:

$$H \mapsto T, \qquad T \mapsto H. \tag{1.4}$$

Now the system alternates forever,

$$H \mapsto T \mapsto H \mapsto T \mapsto \cdots . \tag{1.5}$$

These are already deterministic laws in the strong classical sense: if we know the state at one instant, then we know the entire future sequence.

What matters here is not the coin. What matters is that a classical law is being represented as motion in phase space. If we know where we are, we follow the arrow.

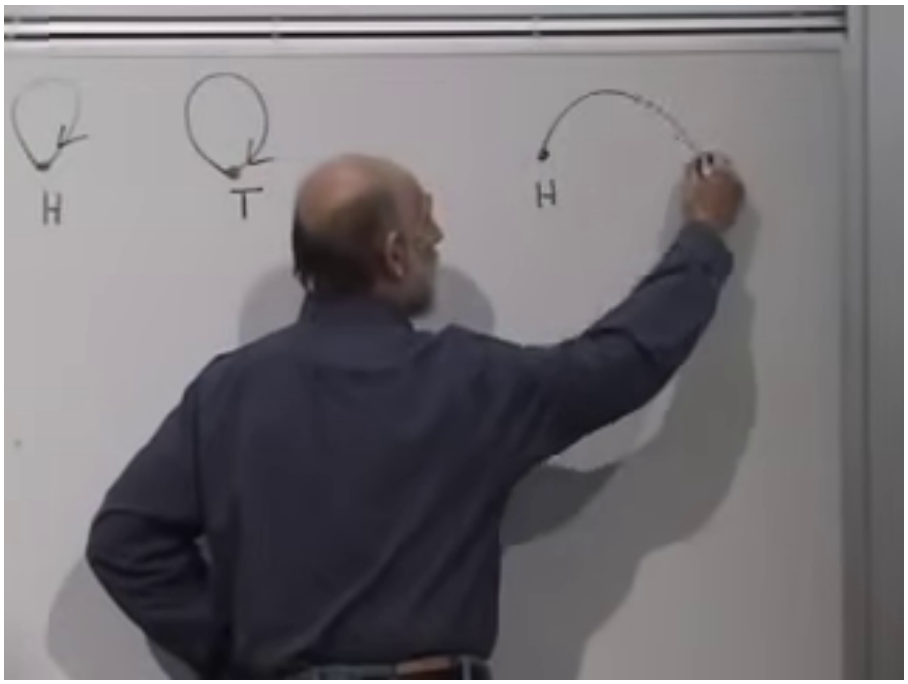


Figure 1.1: Susskind's board as the lecture passes from the trivial self-loop law to the alternating two-state law. The frame directly supports the self-loops and the arrow from H toward T; the reverse arrow is fixed by the transcript.

Figure 1.2: Clean redraw of the two elementary laws on the two-state phase space.

1.4 Allowed and Forbidden Laws in Finite State Spaces

Now the lecture pivots from examples to a criterion. It is no longer enough to draw some arrows. We must ask which arrow diagrams are actually allowed by classical physics.

The first generalization is from two states to six. A die has six faces, so we may imagine six states labelled $1, \ldots, 6$. One possible law is the six-cycle

$$1 \mapsto 2, \qquad 2 \mapsto 3, \qquad 3 \mapsto 4, \qquad 4 \mapsto 5, \qquad 5 \mapsto 6, \qquad 6 \mapsto 1. \tag{1.6}$$

Another acceptable law may break the six states into several disconnected cycles, for example

$$1 \mapsto 1, \qquad 2 \mapsto 3, \qquad 3 \mapsto 4, \qquad 4 \mapsto 2, \qquad 5 \mapsto 6, \qquad 6 \mapsto 5. \tag{1.7}$$

In both cases the rule is deterministic into the future. From each state there is exactly one way to go next.

But classical physics, as the lecture insists, is not only deterministic into the future. It is also deterministic into the past. We must be able to reconstruct where the system came from.

A forbidden example makes the issue plain:

$$1 \mapsto 2, \qquad 2 \mapsto 3, \qquad 3 \mapsto 2.$$

Here the future is unique, but the past is not. If we find ourselves at 2, we cannot tell whether we came from 1 or from 3. Reversing the arrows produces the opposite pathology: then the past is unique but the future is ambiguous.

future unique, past ambiguous past unique, future ambiguous

Figure 1.3: Two forbidden three-state laws. In the first the past is not unique; in the second the future is not unique.

Proposition 1.3. *On a finite phase space Γ, an allowed classical law is a bijection*

$$\sigma : \Gamma \to \Gamma, \tag{1.8}$$

or equivalently a rule with exactly one incoming arrow and exactly one outgoing arrow at each state.

Proof. If a state had two outgoing arrows, the future would not be unique. If it had no outgoing arrow, the system would have no next step. If a state had two incoming arrows, the past would not be unique. If it had no incoming arrow, the motion could not be traced backward through that state. Thus determinism into the future and determinism into the past together require one in-arrow and one out-arrow at every state. On a finite set that is precisely the condition that the update rule be bijective. □

This is the first real mathematical spine of the lecture. A classical finite-state law is a permutation of phase space.

1.4.1 Question & Answer

Question. Why must an allowed classical law have one arrow in and one arrow out at every state?

Answer. Because classical evolution is being characterized here as unique both forward and backward in time. One out-arrow tells us where we go next. One in-arrow tells us where we came from. If either uniqueness fails, information has been lost, and the law no longer has the classical character Susskind is emphasizing.

1.5 Infinite State Spaces, Disconnected Components, and Conservation as Memory

The lecture then removes the assumption that phase space must be finite. Suppose the states are indexed by all integers,

$$\ldots, -2, -1, 0, 1, 2, \ldots,$$

with the law

$$n \mapsto n + 1. \tag{1.9}$$

Each state still has one in-arrow and one out-arrow, but now the trajectory does not cycle through a finite list. It simply marches onward forever.

This already tells us that the finite examples were not important because they were finite. What mattered was the structure of the map on phase space.

The next step is subtler. Phase space may break up into disconnected pieces. One component may be an infinite chain, another a closed cycle, another a fixed point. If the system starts in one component, it never wanders into another. That fact can be rephrased as a conservation law.

This is not yet the conservation of energy or momentum in its familiar form. It is a stripped-down model of the idea of conservation. We may assign a label C to each connected component, for example

$$C = +1 \quad \text{on one component}, \qquad C = -1 \quad \text{on another}, \tag{1.10}$$

and then along the motion we have

$$C_{n+1} = C_n. \tag{1.11}$$

The conserved quantity is simply a memory of where the system started. It never changes because the dynamics never mixes the disconnected parts of phase space.

This is why the lecture identifies reversibility with information conservation. If we know the present state and can reconstruct both the future and the past, then no information about the trajectory has been lost. If two distinct histories collapse into the same present state, information has been destroyed.

At about this point the lecture inserts a valuable warning. Suppose the next coin outcome depended on the previous two entries instead of only the present one. Then the labels H and T would not really be the states after all. The same warning is about to return in continuous form: if position alone does not determine what happens next, then position alone is not the state.

1.6 Continuous Motion and Why Position Alone Is Not Enough

Only after that discrete groundwork does Susskind turn to the real world. He asks for the continuous analogue of the coin question: if a particle moves along a line, is its position enough to determine what happens next?

1.6.1 Question & Answer

Question. Is knowing the position of a particle enough to know what happens next?

Answer. No. We must also know how fast the particle is moving, and in which direction. Position alone is not a complete state. Velocity must be included.

Thus the phase space of a particle on a line is not the line of positions alone. It is the two-dimensional space of pairs

$$(x, v), \qquad v = \dot{x}. \tag{1.12}$$

The horizontal axis is position, the vertical axis is velocity. Positive velocity means motion to the right; negative velocity means motion to the left.

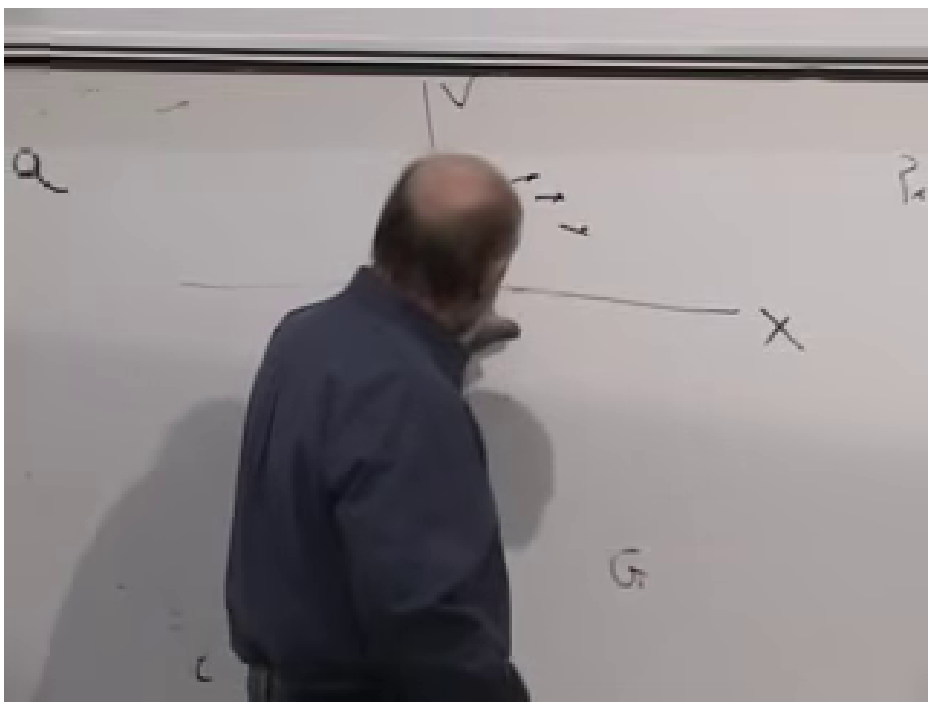


Figure 1.4: Susskind's board near the end of the lecture. The frame supports a local flow picture: a position line, a marked point x, and small arrows indicating the next direction of motion.

The lecture then fills in the meaning of those arrows. If the velocity is zero, the position does not change at the next instant. If the velocity is positive, the next position lies to the right. If the positive velocity is larger, the motion to the right in a fixed time interval is larger. If the velocity is negative, the particle moves to the left.

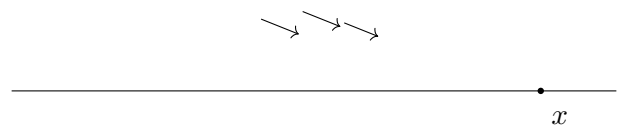


Figure 1.5: A sparse redraw of the local flow sketch. The board frame supports this geometry; the fuller phase-space interpretation comes from the transcript.

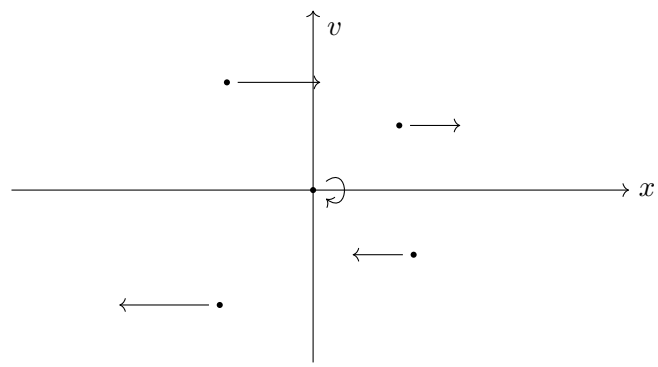


Figure 1.6: Transcript-backed reconstruction of the phase-space picture for motion on a line. Above the x-axis the arrows point right; below it they point left; on the axis itself the next position is unchanged.

The point is no longer merely pictorial. A point in phase space now carries two pieces of information. This is the continuous counterpart of the earlier warning about the coin: whatever information is needed to determine the next step must be built into the definition of state.

The lecture also pauses here over a practical qualification. Classical equations are deterministic in principle, but they are not infinitely predictive in practice. Initial conditions are given by real numbers, and real numbers are never known with infinite precision. Over short intervals the uncertainty may remain small; over longer intervals it may grow, sometimes violently, as in chaotic systems. But the conceptual structure remains the same: for any fixed interval of time, sufficiently precise initial data would, in principle, determine the motion over that interval.

1.7 First-Order Versus Second-Order Equations, and the Return to Two-Step Memory

The next step is to translate the phase-space lesson into the structure of differential equations. The selected board frames do not stabilize these formulas visually, so we write them in the standard notation used throughout the lecture.

A first-order equation contains only first time derivatives. A second-order equation contains accelerations as well:

$$\dot{x} = \frac{dx}{dt}, \qquad \ddot{x} = \frac{d^2x}{dt^2}.$$

To make the logical point as sharply as possible, Susskind first invents a fake law of motion:

$$F(x) = m\dot{x}. \tag{1.13}$$

Assume that the force depends only on the position x. Then if we know x, we know $F(x)$, and the equation immediately tells us $\dot{x}$. Position alone would be enough to determine the next step.

The lecture then differentiates once more:

$$\frac{dF}{dt} = m\ddot{x}, \tag{1.14}$$

$$\frac{dF}{dt} = \frac{dF}{dx}\frac{dx}{dt} = \frac{dF}{dx}\,\dot{x}. \tag{1.15}$$

Since $\dot{x}$ was already known from the first-order equation, $\ddot{x}$ is also determined. Repeating the same idea would determine higher derivatives as well. So a first-order law of this sort would require only one state variable.

But Newton's law does not have that form. Newton's law is

$$F(x) = m\ddot{x}. \tag{1.16}$$

Now if we know the position, we know the force and hence the acceleration. But there is still no equation that tells us the velocity. Therefore position alone does not specify the state. We must supply the velocity as additional information. That is why the phase space is two-dimensional.

This is the same lesson we learned from the coin diagrams, now stated in the language of calculus: the order of the equation measures how much information a state must contain.

1.7.1 Question & Answer

Question. What if the next state depends on the previous two configurations?

Answer. Then a single H or T is not a complete state. The state must be enlarged so that it includes both pieces of information. The law may still be deterministic and reversible, but only on the enlarged phase space.

The lecture returns to the coin example precisely to sharpen this point. If the law depends on the last two entries, then the proper phase space is

$$\Gamma_2 = \{HH, HT, TH, TT\}. \tag{1.17}$$

A clean way to write such a law is to say that the next symbol is a function of the previous pair:

$$(a, b) \mapsto (b, f(a, b)).$$

One reversible completion consistent with the spirit of the classroom discussion is

$$f(H, H) = H, \qquad f(H, T) = T, \qquad f(T, H) = T, \qquad f(T, T) = H,$$

which induces the update rule

$$HH \mapsto HH, \qquad HT \mapsto TT, \qquad TT \mapsto TH, \qquad TH \mapsto HT. \tag{1.18}$$

This is now a deterministic reversible law on four states. The difficulty was never that the law itself was impossible. The difficulty was that the original state space $\{H, T\}$ was too small.

Once that is clear, the higher-order moral follows immediately. If classical mechanics had required positions, velocities, and accelerations as independent initial data, then the equations of motion would have been third order in time and the phase space would have had to include

$$(x, \dot{x}, \ddot{x}).$$

Experiment tells us that ordinary classical mechanics does not work that way. It stops at positions and velocities. For a single particle in three-dimensional space, the phase space therefore has six coordinates: three positions and three velocities.

1.8 Summary

The lecture begins with an intentionally oversized claim and then earns it by working downward to the simplest possible examples. We first learn to think of dynamics as motion in phase space; then we learn that a state is exactly the information needed to know what happens next; then we discover that classical evolution has the special structure of determinism both into the future and into the past.

In the discrete toy world this structure appears as one in-arrow and one out-arrow at every point. In the continuous world it appears as the necessity of using (x, v) rather than x alone, and in the equations of motion it reappears as the difference between a first-order law and Newton's actual second-order law. The lecture's unifying theme is already clear in the first hour: classical mechanics is the study of lawful evolution as conservation of information.

Chapter 2

From Initial Data to Least Action

These notes follow Leonard Susskind's second lecture on classical mechanics, as preserved in the course recording curated by LazyingArt LLC and Video2Book. The lecture opens as review, but it is not idle review. We begin with the question of initial conditions, use a deliberately false law to see what the order of a differential equation means, recover Newton's actual structure by contrast, review energy and momentum in that Newtonian setting, and then turn toward the deeper formulation in which one studies whole trajectories rather than local steps along them.

2.1 First-Order and Second-Order Laws

Susskind begins with one-dimensional motion and a false law of physics. The point is not to rescue Aristotle; it is to understand what a first-order equation would imply if nature had chosen one. The law is

$$F = m\dot{x}, \tag{2.1}$$

or, if the force is known as a function of position,

$$F(x) = m\dot{x}. \tag{2.2}$$

This law would say that in order to keep an object moving one must continuously exert a force on it. Remove the force, and the object comes immediately to rest. That is not our world. But mathematically the law is consistent, and that is enough for the comparison.

Suppose we know the position x at some instant. Then we know the force $F(x)$. But the equation itself then tells us the velocity $\dot{x}$. So for a first-order law of this kind, position already fixes the first time derivative. The natural question is whether it also fixes the higher derivatives. It does.

The force changes with time only because the particle moves through regions where the force differs. Therefore

$$\frac{dF}{dt} = \frac{dF}{dx}\,\dot{x}. \tag{2.3}$$

Differentiating $F(x) = m\dot{x}$ with respect to time gives

$$m\ddot{x} = \frac{dF}{dt} = \frac{dF}{dx}\,\dot{x}. \tag{2.4}$$

So once the position is known, the velocity is known, and once the velocity is known, the acceleration is known. Differentiate once more:

$$m\,\dddot{x} = \frac{d}{dt}\left(\frac{dF}{dx}\,\dot{x}\right) \tag{2.5}$$

$$= \frac{d^2F}{dx^2}\,\dot{x}^{\,2} + \frac{dF}{dx}\,\ddot{x}. \tag{2.6}$$

The jerk is now determined as well. The pattern is obvious enough that Susskind is content to stop there: one can continue indefinitely, and each new time derivative is fixed by the previous ones and by derivatives of the force law. A first-order theory closes from position alone.

Now rerun the same question for Newton:

$$F(x) = m\ddot{x}. \tag{2.7}$$

If we know the position, then we know the force, and therefore we know the acceleration. But the equation says nothing whatever about the velocity. The obstruction is immediate and decisive. A second-order equation does not propagate from position alone. To get started we must specify both

$$x(t_0), \qquad \dot{x}(t_0). \tag{2.8}$$

Only then do we know enough to determine the acceleration, and from there the higher derivatives.

This is the first real lesson of the lecture: Newtonian mechanics is second order in time, and so its initial data are positions and velocities. For many particles the statement is the same in larger dimension: every coordinate and every corresponding velocity component must be supplied.

2.1.1 Question & Answer

Question. Why does Newton need both position and velocity while Aristotle's law did not?

Answer. Because the issue is the order of the differential equation. In the first-order law $F(x) = m\dot{x}$, knowing the position immediately gives the first derivative. In Newton's law $F(x) = m\ddot{x}$, knowing the position gives only the second derivative. The first derivative remains independent, so velocity must be provided as part of the initial data.

2.2 Conservative Forces and Energy Conservation

Having reestablished the initial-data problem, Susskind turns to a quick review of energy conservation. He marks the level of the argument very clearly: the proof is elementary, but the principle is deeper than the proof. Energy conservation survives in electrodynamics, relativity, and quantum mechanics. Here we are only checking that Newton's equations reproduce it in the ordinary conservative case.

He first reminds us that not every force law conserves the mechanical sum of kinetic and potential energy. If we model friction crudely, the kinetic energy appears simply to disappear. Likewise, if some force kept pushing a particle around a circle in the same direction, the particle would come back to the same position with greater speed, so the kinetic energy would increase while the position, and hence any ordinary potential energy, returned to the same value. Such examples are a warning: the familiar energy law is not automatic.

The forces that occur in nature, at least in the Newtonian particle examples under discussion, are conservative. For a single particle they take the form

$$F_i = -\frac{\partial U}{\partial x_i}, \qquad \vec{F} = -\nabla U. \tag{2.9}$$

The minus sign is the content: force points in the direction of decreasing potential energy.

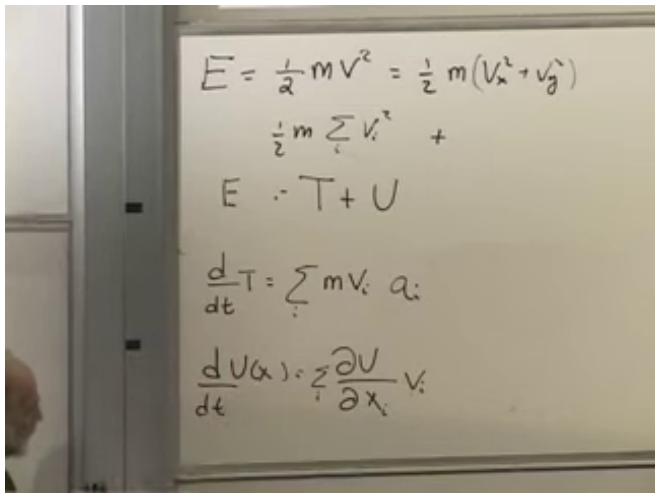


Figure 2.1: Energy definition and separate time-derivative terms. The top handwritten symbol is best read as T, not E; the transcript and the later cancellation make that normalization clear.

Now define the kinetic and total energies:

$$T = \frac{1}{2}mv^2 = \frac{1}{2}m\sum_i v_i^2, \qquad E = T + U. \tag{2.10}$$

The board in the lecture is worth preserving because it shows the order of the argument. First one writes down T, then $E = T + U$, then the separate derivatives of T and U. Only afterward is the cancellation exposed.

The derivative of the kinetic energy is

$$\frac{dT}{dt} = \frac{d}{dt}\left(\frac{1}{2}m\sum_i v_i^2\right) \tag{2.11}$$

$$= \sum_i mv_i\frac{dv_i}{dt} \tag{2.12}$$

$$= \sum_i mv_i a_i. \tag{2.13}$$

The derivative of the potential energy follows from the chain rule:

$$\frac{dU}{dt} = \sum_i \frac{\partial U}{\partial x_i}\frac{dx_i}{dt} \tag{2.14}$$

$$= \sum_i \frac{\partial U}{\partial x_i}v_i. \tag{2.15}$$

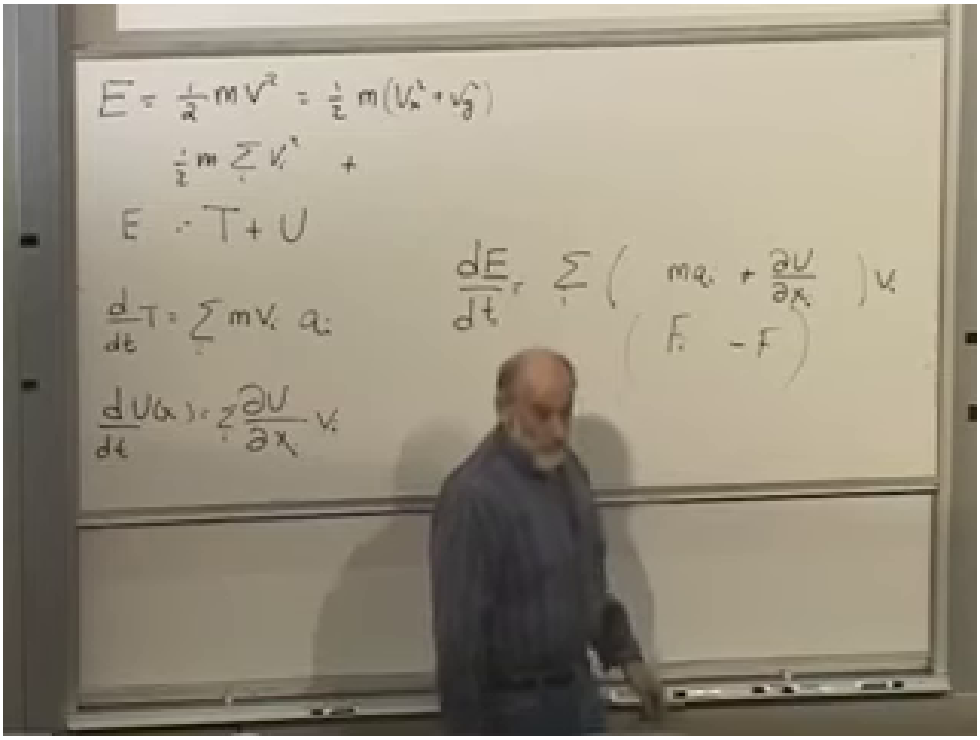


Figure 2.2: Combined time derivative of the total energy and the force-minus-force cancellation.

Now add the two pieces:

$$\frac{dE}{dt} = \frac{dT}{dt} + \frac{dU}{dt} \tag{2.16}$$

$$= \sum_i \left(ma_i + \frac{\partial U}{\partial x_i} \right) v_i. \tag{2.17}$$

At this point the grouping on the board becomes transparent. We write $ma_i = F_i$, while the conservative-force law says

$$\frac{\partial U}{\partial x_i} = -F_i. \tag{2.18}$$

Therefore

$$\frac{dE}{dt} = \sum_i (F_i - F_i) v_i = 0. \tag{2.19}$$

That is the conservation of energy.

The lecture stresses not just the result but the rhythm of the proof. The two derivatives are built separately; only then are they collected into a bracket; and only then do we read the bracket as force minus force. This is review, but it is review with purpose. Already one feels that the conservation law is telling us something more organized than a mere equation of motion.

The same argument extends immediately to many-particle systems. The index i can stand not only for spatial directions but also for different particles. As long as the force on each coordinate is minus the derivative of the potential with respect to that coordinate, the total energy of the whole system is conserved.

2.3 Force as the Time Derivative of Momentum

Susskind now says, in effect, let us come to momentum. Again the mood is review, but the review is widening. Newton's law is rewritten in a slightly different form,

$$F_i = m\frac{dv_i}{dt}. \tag{2.20}$$

Since the mass is constant, we may pull it inside the derivative:

$$F = \frac{d}{dt}(mv). \tag{2.21}$$

And once we give mv a name,

$$\vec{P} = m\vec{v}, \tag{2.22}$$

Newton's law becomes

$$\vec{F} = \frac{d\vec{P}}{dt}. \tag{2.23}$$

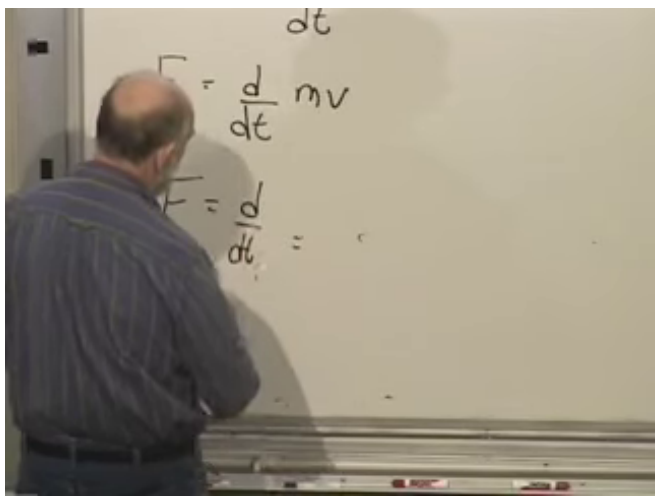


Figure 2.3: Newton's law being rewritten in momentum form. This is a transition frame rather than a finished formula, but it preserves the live progression of the board.

The board hierarchy in the lecture is worth reproducing in clean form:

$$F_i = m\frac{dv_i}{dt}, \tag{2.24}$$

$$F = \frac{d}{dt}(mv), \tag{2.25}$$

$$\vec{F} = \frac{d\vec{P}}{dt}. \tag{2.26}$$

Before going further, the lecture pauses for a sharp question: what assumptions about time are already buried in this Newtonian discussion? The answer is explicit. We are assuming a universal synchronized time, the same at every point in space. We are also assuming, for the moment, that the force law has no explicit time dependence. This is the most naive Newtonian picture, and the lecture wants that simplicity stated out loud before momentum conservation is derived within it.

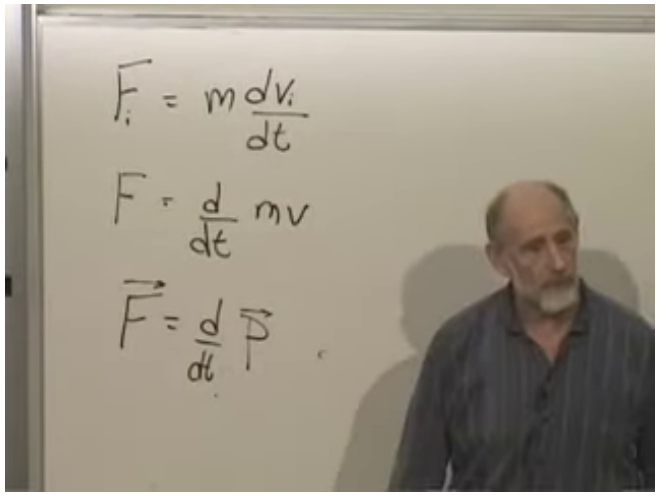


Figure 2.4: Component and vector forms of force and momentum.

Now take a closed system of particles. The force on each particle is the sum of forces due to all the others. For three particles,

$$\vec{F}_1 = \vec{F}_{12} + \vec{F}_{13}, \tag{2.27}$$
$$\vec{F}_2 = \vec{F}_{21} + \vec{F}_{23}, \tag{2.28}$$
$$\vec{F}_3 = \vec{F}_{31} + \vec{F}_{32}. \tag{2.29}$$

The notation is important: $\vec{F}_{12}$ means the force on particle 1 due to particle 2. Newton's third law then says

$$\vec{F}_{12} = -\vec{F}_{21}, \qquad \vec{F}_{13} = -\vec{F}_{31}, \qquad \vec{F}_{32} = -\vec{F}_{23}. \tag{2.30}$$

The last of these corrects a small slip in the transcripted cancellation line; the intended pair is clearly $\vec{F}_{32}$ and $\vec{F}_{23}$.

Since $\dot{\vec{P}}_i = \vec{F}_i$, we have

$$\frac{d\vec{P}_1}{dt} = \vec{F}_{12} + \vec{F}_{13}, \tag{2.31}$$
$$\frac{d\vec{P}_2}{dt} = \vec{F}_{21} + \vec{F}_{23}, \tag{2.32}$$
$$\frac{d\vec{P}_3}{dt} = \vec{F}_{31} + \vec{F}_{32}. \tag{2.33}$$

Add them:

$$\frac{d}{dt}(\vec{P}_1 + \vec{P}_2 + \vec{P}_3) = \vec{F}_{12} + \vec{F}_{13} + \vec{F}_{21} + \vec{F}_{23} + \vec{F}_{31} + \vec{F}_{32} \tag{2.34}$$
$$= 0. \tag{2.35}$$

Hence

$$\frac{d\vec{P}_{\text{tot}}}{dt} = 0. \tag{2.36}$$

So the total momentum of a closed system is conserved.

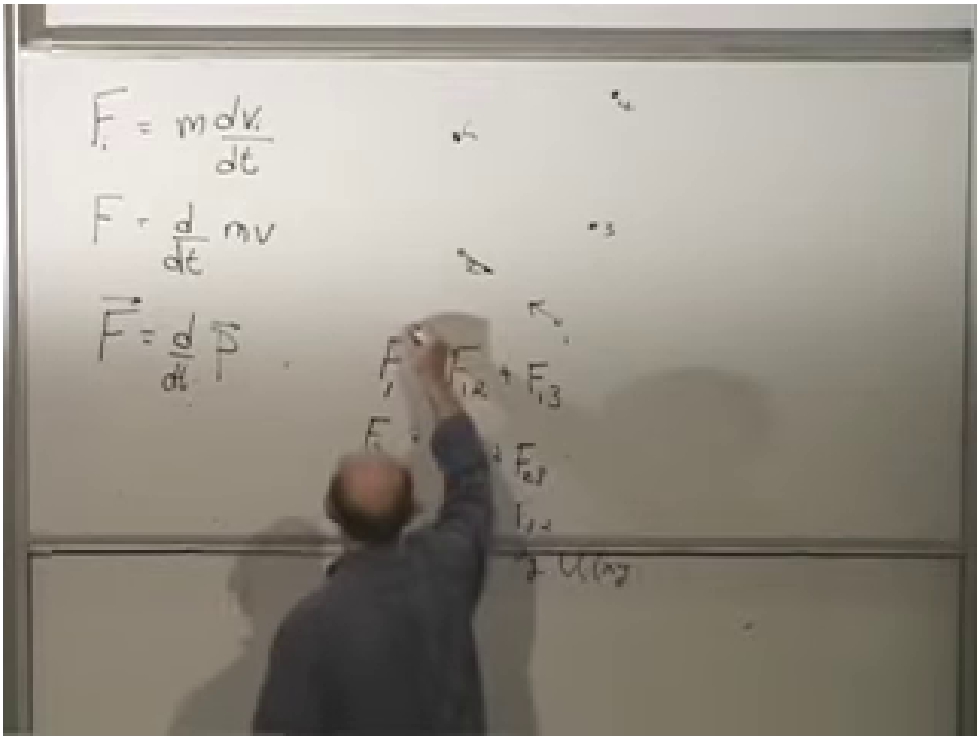


Figure 2.5: Pairwise force sketch for momentum conservation. The exact subscripts in the board drawing are not all legible, so the transcript controls the notation while the image preserves the layout and vector logic.

The lecture immediately widens the scope. Momentum conservation is far more general than this little Newtonian example suggests. It applies to light, to quantum mechanics, to relativity. The derivation we have given uses Newtonian assumptions, but the law itself is deeper.

2.3.1 Question & Answer

Question. Are action and reaction instantaneous, and which part of this argument is specifically Newtonian?

Answer. In Newtonian mechanics, yes: forces are taken to act instantaneously, because time is assumed universal and signal propagation effectively infinite. That is the specifically Newtonian ingredient in the derivation. But the conservation of momentum itself survives when relativity removes instantaneous action at a distance. So the argument is Newtonian; the law is more general.

2.4 A Mathematical Warm-Up: Stationary Points

Only now does the lecture deliberately slow down and say: before we go on to the deep law of classical physics, we need a little mathematics. Not heavy mathematics, but just enough to put everyone back on the same playing field.

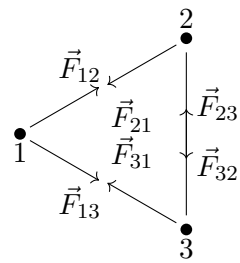


Figure 2.6: A cleaned pairwise-force diagram clarifying the cancellation of internal forces.

Take an ordinary function $f(y)$ of one variable. If we look for a local minimum, the first condition is that the slope vanish:

$$\frac{df}{dy} = 0. \tag{2.37}$$

This condition is not sufficient to distinguish minima from maxima, and it also includes inflection points where the graph is momentarily flat. So the mathematically safer word is not minimum but stationary point. The function does not change to first order under a small change of the variable.

Susskind makes that point explicitly because the principle of least action is really the principle of stationary action. He says he will mostly keep the simpler phrase "least action," but only after flagging the distinction.

For several variables the first-order condition simply splits into one equation per direction:

$$\frac{\partial f}{\partial x_i} = 0 \qquad \text{for every } i. \tag{2.38}$$

If the function is stationary, it must be stationary no matter which direction one moves away from the point.

The lecture then works a concrete two-variable example,

$$f(x, y) = x^2 + y^2 + 3x + 2y + xy + 7. \tag{2.39}$$

The stationary-point equations are

$$\frac{\partial f}{\partial x} = 2x + y + 3 = 0, \tag{2.40}$$

$$\frac{\partial f}{\partial y} = x + 2y + 2 = 0. \tag{2.41}$$

The constant 7 disappears, as it should: an overall vertical shift changes the height of the function but not the location of the stationary point. Solving the equations carefully gives

$$x = -\frac{4}{3}, \qquad y = -\frac{1}{3}. \tag{2.42}$$

The lecture's spoken numbers are somewhat casual, but the equations themselves are stable, and the method is the point.

The transition is now ready. We know how to locate stationary points of ordinary functions. The next step is to study quantities that depend not on finitely many variables but on whole functions, whole curves, whole trajectories.

2.4.1 Question & Answer

Question. Is the principle really least action, or stationary action?

Answer. Strictly speaking, stationary action. What must vanish is the first-order change in the quantity under a small deformation of the path. That condition includes minima, but it can also include maxima or other stationary cases. In the ordinary mechanical examples at hand, one often does have a minimum, and the lecture is content to speak that way, but the more accurate phrase is stationary action.

2.5 Local Motion, Whole Paths, and Functionals

At this point the lecture does something subtle. Before giving geometric examples, it restates the basic problem of mechanics itself. A mechanical system is described by generalized coordinates

$$q_1,\ q_2,\ \ldots,\ q_n, \tag{2.43}$$

which may be Cartesian coordinates, angles, orientations, latitude and longitude, or any other variables that locate the system in configuration space. The full motion is then the collection of functions

$$q_1(t),\ q_2(t),\ \ldots,\ q_n(t). \tag{2.44}$$

That collection is the generalized trajectory.

In the Newtonian formulation, to predict the future we must know the generalized coordinates and their time derivatives at one instant:

$$q_i(t_0), \qquad \dot{q}_i(t_0). \tag{2.45}$$

Newton's equations are local equations along the trajectory. If we know where we are and how we are moving, the equations tell us the acceleration, and so they tell us how to proceed from one point of the trajectory to the next. One does not need knowledge of the whole curve in advance.

The principle of least action will formulate the same mechanics differently. Instead of the position and velocity at one time, we specify the positions at two times, the beginning and the end of the interval. That is a different but equally sufficient way to pose the problem. Susskind emphasizes this with the simple thought experiment of an object whose initial position is known and whose final position after one second is also known. The true trajectory is the one that connects those two endpoints in the allotted time.

Now the geometric examples can do their work. Consider first the shortest curve between two fixed points. Let $y(x)$ describe the curve. For a small segment,

$$ds^2 = dx^2 + dy^2, \tag{2.46}$$

so

$$ds = dx\sqrt{1 + \left(\frac{dy}{dx}\right)^2}. \tag{2.47}$$

Integrating from x_1 to x_2, the total length is

$$s_{12}[y] = \int_{x_1}^{x_2} \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx. \tag{2.48}$$

This is not a function of a few variables. It is a rule that takes an entire function $y(x)$ and returns a number. That is what the lecture now names: a functional.

The lecture insists on both a global and a local viewpoint. Globally, we minimize the total length of the whole curve. Locally, if we focus on a tiny segment between two nearby pinned points, the rule is simply: go straight ahead. The local differential-equation point of view and the global minimization point of view are different ways of expressing the same geometry.

There is also an instructive false shortcut. One might think that since the integrand contains $1 + (dy/dx)^2$, the shortest path should come from setting $dy/dx = 0$. But that would only produce a horizontal line, and a horizontal line will not generally pass through the fixed endpoints. The endpoint condition is part of the problem; one cannot minimize the integrand pointwise while ignoring the global constraint.

The optical example is even closer to what is coming in mechanics. Suppose the speed of light depends on height, $c = c(y)$, as in a layered medium. Then

$$c(y) = \frac{ds}{dt}, \qquad dt = \frac{ds}{c(y)}. \tag{2.49}$$

Substituting the expression for ds,

$$dt = \frac{\sqrt{1 + \left(\frac{dy}{dx}\right)^2}}{c(y)}\, dx. \tag{2.50}$$

Hence the total travel time is

$$t_{12}[y] = \int_{x_1}^{x_2} \frac{\sqrt{1 + \left(\frac{dy}{dx}\right)^2}}{c(y)}\, dx. \tag{2.51}$$

The true light ray is the curve $y(x)$ that makes this time stationary.

The lecture adds one more example in passing: a hanging chain of fixed length chooses the curve that minimizes its potential energy. The exact catenary formula is not derived here, but the point is the same. Physics repeatedly asks us to minimize quantities that depend on whole curves, not just on finitely many coordinates. That is the calculus of variations.

2.6 The Action

Only after this entire runway does Susskind finally say: now let me tell you what the action is. He presents it as something surprising and, even after decades, still surprising. The pedagogical rhythm matters. We are first invited to guess the wrong answer.

Take one particle moving in one dimension, with trajectory $x(t)$. Since the action is to be built from kinetic and potential energy and integrated over time, the naive guess would be

$$\int_{t_0}^{t_1} (T + U)\, dt. \tag{2.52}$$

That is the expected guess. But it is wrong.

The correct quantity is

$$A[x] = \int_{t_0}^{t_1} (T - U)\, dt = \int_{t_0}^{t_1} \left(\frac{1}{2}m\dot{x}^2 - U(x)\right)\, dt. \tag{2.53}$$

This is the action. The integrand is called the Lagrangian:

$$L = T - U. \tag{2.54}$$

At this point the lecture compares the action directly with the earlier functionals. In the least-distance problem the integrand depended on $y(x)$ and on dy/dx. In the least-time problem it depended on $y(x)$, on dy/dx, and on the local light speed $c(y)$. Here, too, the integrand depends on the function and its first derivative: the position $x(t)$ enters through the potential energy $U(x)$, and the derivative $\dot{x}$ enters through the kinetic energy.

This generalizes immediately. For a mechanical system with generalized coordinates $q_i(t)$,

$$L(q, \dot{q}) = T - U, \qquad A[q] = \int_{t_0}^{t_1} L(q, \dot{q})\, dt. \tag{2.55}$$

In Newtonian particle mechanics the Lagrangian is the difference between kinetic and potential energy. In more general systems the Lagrangian may be a different function of q and $\dot{q}$, but the action principle survives. That is why Susskind calls it the deepest law of classical physics.

The lecture also pauses over notation. Since s has already been used for path length, it is awkward to use s for action. The letter does not matter much, but the caution is sensible: the action is not to be confused with the distance functional we have just finished discussing.

The chapter should end where the lecture ends: at the threshold. The action has been revealed, but not yet varied. We have not derived the equations of motion from it. That task is deferred to the next lecture, where the analogy with optics will be completed and the least-action principle will be turned back into differential equations.

2.6.1 Question & Answer

Question. Why is the action built from kinetic minus potential energy rather than from the total energy?

Answer. At the level of this lecture, the honest answer is that $T - U$ is the combination that reproduces Newton's equations when the variational problem is worked out. Susskind explicitly postpones a deeper motivation and prefers first to examine the consequences. The minus sign is not ornamental. Once the potential is defined by $F = -dU/dx$, the sign in the action determines the sign in the resulting equation of motion. The naive integral of $T + U$ does not yield the Newtonian law in the form we want; the integral of $T - U$ does.

2.7 Summary

The lecture begins with a question about initial conditions and ends with the action, but the argument unfolds in a very deliberate sequence. A false first-order law shows what it would mean for position alone to determine the whole motion. Newton's second-order law then shows why velocity must be supplied independently. Energy conservation is reviewed by building its ingredients separately and letting them cancel only at the end. Momentum conservation is derived from pairwise action and reaction, then immediately recognized as deeper than the Newtonian assumptions used in the derivation.

Only after that review does the lecture broaden its scope. We pass from stationary points of ordinary functions to functionals of whole curves, from local differential rules to global variational problems, and from fixed initial data to fixed endpoints. The payoff is the final formula

$$A[q] = \int_{t_0}^{t_1} L(q, \dot{q})\, dt, \qquad L = T - U \tag{2.56}$$

for Newtonian particle mechanics. The next step, and the next lecture's task, is to show how this global principle reproduces the local equations of motion.

Chapter 3

Least Action, Euler–Lagrange, and Symmetry

We begin, as the lecture begins, at a surprisingly large scale. The subject is not merely the motion of one particle, nor even mechanics taken narrowly, but the claim that the laws of physics share a common form. From that wide opening the lecture narrows deliberately: first two small mathematical tools, then the variational derivation itself, and only after that the familiar force laws and conservation laws. The guiding idea throughout is that a global statement about an entire history can be converted into a local equation along the path. The retained board figures come from Leonard Susskind's lecture, with visual curation by LazyingArt LLC and Video2Book, and they are kept only where they witness notation, layout, or geometry that genuinely matters for the mathematics.

3.1 The Common Form of Physical Law

The lecture does not start with $F = ma$. It starts with the broader claim that the laws of physics have a common form. There are mathematical truths, and there are theorems of probability theory, but these are not yet the laws of physics in the specific sense Susskind wants. The laws he has in mind are the dynamical laws: the motion of particles, gravity, electromagnetism, the motion of charged particles, and more generally the machinery by which physical systems evolve.

His claim is that all of these classical laws can be written in a common structural form, and that form is the principle of least action. Even thermodynamics, which is not usually written this way, is placed under the same umbrella: its laws are statistical laws of many degrees of freedom whose underlying microscopic dynamics are themselves governed by an action principle.

This opening matters for the tone of the chapter. Least action is not presented as a decorative reformulation of Newtonian mechanics. It is presented as a common language of physics. Only after making that claim does the lecture step back and say, in effect, before we touch the main argument we need two little theorems.

3.2 Two Mathematical Tools Before the Main Derivation

The first theorem is integration by parts. The transcript is garbled in places here, so we regularize the notation into its standard form while keeping the lecture's actual purpose in view.

We begin with the elementary fact that the definite integral of a derivative is the difference of the function at the endpoints:

$$\int_{t_1}^{t_2} \dot{F}(t)\,dt = F(t_2) - F(t_1). \tag{3.1}$$

Now let $F(t) = f(t)g(t)$. The product rule gives

$$\frac{d}{dt}\big(f(t)g(t)\big) = \dot{f}(t)\,g(t) + f(t)\,\dot{g}(t). \tag{3.2}$$

Integrating from t_1 to t_2, we obtain

$$\int_{t_1}^{t_2} \big(\dot{f}\,g + f\,\dot{g}\big)\,dt = [fg]_{t_1}^{t_2}, \tag{3.3}$$

and therefore

$$\int_{t_1}^{t_2} \dot{f}\,g\,dt = [fg]_{t_1}^{t_2} - \int_{t_1}^{t_2} f\,\dot{g}\,dt. \tag{3.4}$$

This is the form we shall use later. It lets us move the dot from one factor to the other, but only at the price of a minus sign and a boundary term.

3.2.1 Question & Answer

Why is there a dt in the integral? This is exactly the sort of interruption the lecture wants to preserve. The answer is simple and worth stating plainly: an integral must always be an integral with respect to something. In this discussion the independent variable is time, so one must write dt. The expression

$$\int \dot{f}(t)$$

is incomplete notation; the intended expression is

$$\int \dot{f}(t)\,dt.$$

The point is elementary, but the lecture uses it to slow the argument down and make the notation explicit before the harder variational step.

The special case we need repeatedly is the case in which one factor vanishes at both endpoints. Then the boundary term drops out and the formula becomes

$$\int_{t_1}^{t_2} \dot{f}\,g\,dt = -\int_{t_1}^{t_2} f\,\dot{g}\,dt. \tag{3.5}$$

This is exactly the version that will be used when the variation of the path is fixed to vanish at the initial and final times.

The second tool is even smaller, but in the derivation it is decisive.

Proposition 3.1. *If*

$$\int_{t_1}^{t_2} a(t) f(t)\, dt = 0 \tag{3.6}$$

for arbitrary functions $f(t)$, then $a(t) = 0$ on the interval.

Proof. The lecture's argument is the right one. Choose $f(t)$ to vanish everywhere except in a tiny neighborhood of one point t_*. Then the integral is proportional to the value of $a(t)$ near that point. If the integral must vanish for every such choice of f, then $a(t_*) = 0$. Since the point t_* was arbitrary, $a(t)$ must vanish everywhere. □

Together, these two small pieces of mathematics do one large job. Integration by parts moves the derivative, and the arbitrary-function lemma converts an integral statement into a pointwise one. With those in hand, the lecture says we are ready for the big one.

3.3 Trajectories, Histories, and the Local–Global Tension

A system is described by generalized coordinates

$$q_1(t),\ q_2(t),\ \ldots,\ q_n(t).$$

A trajectory, or a history, is the entire collection of these functions of time. To know the history is to know all the q_i for all times in the interval of interest.

The lecture is careful here about how to picture this. In elementary mathematics the independent variable is usually placed horizontally; in physics we often prefer to imagine time running upward. That pictorial choice is secondary, but the point it serves is not. A history is not a single point. It is a path in time.

There are then two ways to formulate the laws of motion.

The first is local. A local law tells us how to move forward from one small piece of the trajectory to the next. Newton's law $F = ma$ is the example Susskind has in mind. If we know the coordinates and velocities at one instant, the differential equation tells us how the path continues.

The second is global. A global formulation looks at the whole trajectory at once. It assigns to that trajectory a quantity called the action, and the true trajectory is the one for which the action is minimized, or more carefully, stationary, among nearby trajectories with the same endpoints.

This is the tension the lecture wants to resolve: how can a global statement about a whole history be equivalent to a local differential law? The motivating answer is simple. If the true trajectory minimizes the action for the whole interval, it must also minimize the action between neighboring points of the path. Otherwise we could replace that local segment by a better one and lower the total action. So the global principle should imply a local equation. That is exactly what the derivation now does.

3.4 Varying the Path and Deriving the Euler–Lagrange Equation

Let $\hat{q}_i(t)$ denote the true path. We deform it by adding a small multiple of an arbitrary function:

$$q_i(t) = \hat{q}_i(t) + \alpha f_i(t). \tag{3.7}$$

Here α is just a number, a parameter measuring how much we have deformed the path. The functions $f_i(t)$ are otherwise arbitrary, except that they must vanish at the endpoints:

$$f_i(t_1) = f_i(t_2) = 0. \tag{3.8}$$

This keeps the initial and final points fixed while allowing the trajectory to vary in between.

At precisely this point the lecture pauses and asks the right question: what is the action whose minimum we are discussing?

Definition 3.2. For a trajectory $q_i(t)$ between t_1 and t_2, the action is

$$A[q] = \int_{t_1}^{t_2} L(q, \dot{q}, t)\, dt. \tag{3.9}$$

In the mechanical examples of this lecture,

$$L = T - U, \tag{3.10}$$

kinetic energy minus potential energy.

The lecture is also careful not to use $L = T - U$ too early in the derivation. For the moment we let L be any function of the coordinates and velocities. The action is then the sum, all along the path, of these little incremental contributions.

Once the path depends on α, the action becomes a function $A(\alpha)$. The true path is the undeformed one, $\alpha = 0$. Since the true path is the stationary path,

$$\left.\frac{dA}{d\alpha}\right|_{\alpha=0} = 0. \tag{3.11}$$

Now differentiate the varied path with respect to α:

$$\frac{\partial q_i}{\partial \alpha} = f_i(t), \qquad \frac{\partial \dot{q}_i}{\partial \alpha} = \dot{f}_i(t). \tag{3.12}$$

Using the chain rule, we obtain the first variation

$$\frac{dA}{d\alpha} = \int_{t_1}^{t_2} \sum_i \left(\frac{\partial L}{\partial q_i} f_i + \frac{\partial L}{\partial \dot{q}_i} \dot{f}_i \right) dt. \tag{3.13}$$

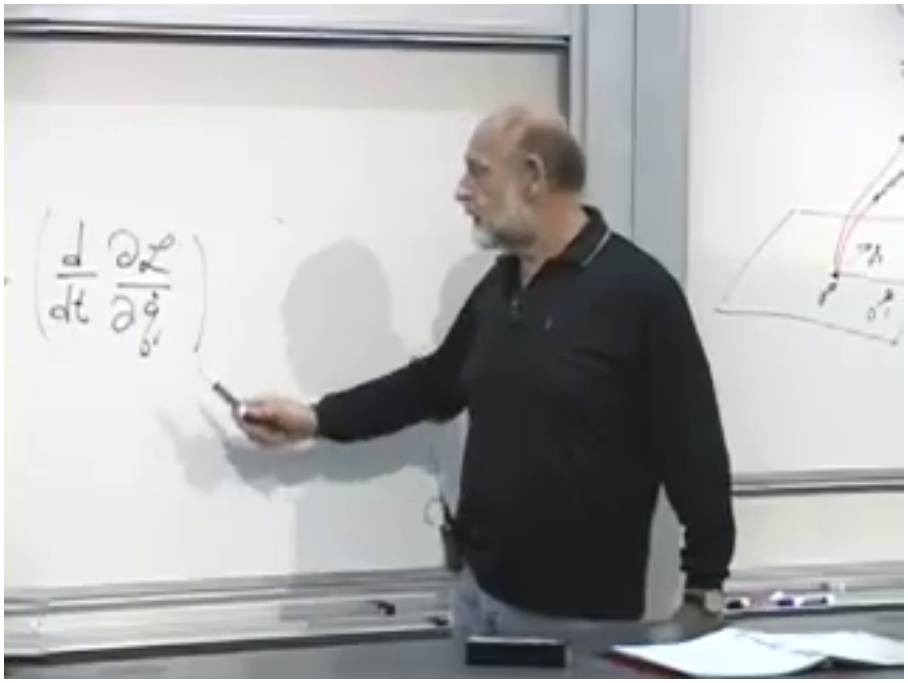
The transcript is noisy at this stage, but the structure is clear: the first term comes from the change in the coordinates, the second from the change in the velocities.

We now use the earlier mathematical tool exactly where it was prepared for use. Integrating the second term by parts,

$$\int_{t_1}^{t_2} \frac{\partial L}{\partial \dot{q}_i} \dot{f}_i\, dt = \left[f_i \frac{\partial L}{\partial \dot{q}_i} \right]_{t_1}^{t_2} - \int_{t_1}^{t_2} f_i \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) dt. \tag{3.14}$$

The boundary term vanishes because every f_i vanishes at t_1 and t_2. Hence

$$\frac{dA}{d\alpha} = \int_{t_1}^{t_2} \sum_i \left[\frac{\partial L}{\partial q_i} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) \right] f_i\, dt. \tag{3.15}$$



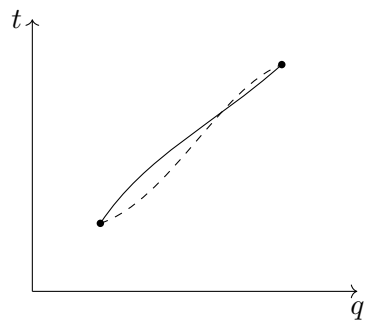


Figure 3.1: The board isolates the time-derivative term in the Euler–Lagrange equation while keeping a nearby trajectory sketch in view. The redraw below is intentionally qualitative: it records only the lecture's local idea of comparing nearby paths between fixed points.

At $\alpha = 0$, stationarity requires this to be zero for every choice of the functions $f_i(t)$. By the arbitrary-function lemma, the bracket itself must vanish pointwise. Therefore

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) = \frac{\partial L}{\partial q_i}. \tag{3.16}$$

This is the Euler–Lagrange equation.

The lecture's central achievement is exactly this one. A global condition on the whole trajectory has been converted into a local equation that holds at each point along the path.

As it appears on the board, in cleaned notation, the central structure is

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i}\right) = \frac{\partial \mathcal{L}}{\partial q_i}. \tag{3.17}$$

In the chapter we use L for the Lagrangian, but the board figure is worth keeping because it witnesses the way Susskind visually separates the algebra from the trajectory picture.

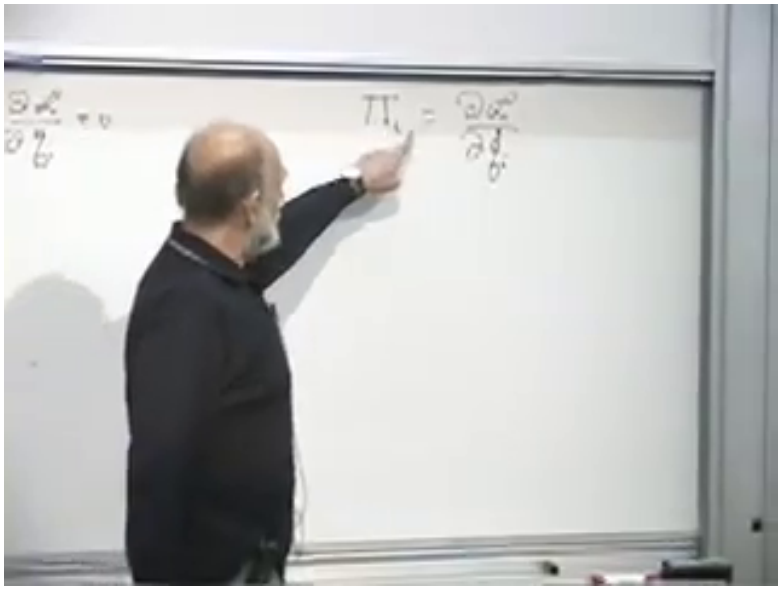


Figure 3.2: The lecture introduces the momentum conjugate to q_i by differentiating the Lagrangian with respect to $\dot{q}_i$. The board symbol can be read as a capital Π_i, but the lecture and standard notation support the lowercase π_i used here.

3.5 What the Euler–Lagrange Equation Says

Once the formal derivation is complete, the lecture immediately turns to language. The new quantities are named only after the structure is in place.

We define

$$\pi_i = \frac{\partial L}{\partial \dot{q}_i}. \tag{3.18}$$

This is the canonical momentum conjugate to q_i. In the board figure the leading symbol is visually a little ambiguous, but both the lecture's language and standard usage support the lowercase π_i, which we keep throughout.

In parallel, the quantity

$$\frac{\partial L}{\partial q_i} \tag{3.19}$$

is called the generalized force. The Euler–Lagrange equation therefore reads

$$\frac{d\pi_i}{dt} = \frac{\partial L}{\partial q_i}. \tag{3.20}$$

Now the equation begins to sound familiar: the time derivative of a momentum equals a force.

3.5.1 Question & Answer

What does the Euler–Lagrange equation say in English? The lecture deliberately postpones this question until after examples, and that postponement is pedagogically right. The first answer is structural: the equation says that a global least-action principle is equivalent to a local differential law along the trajectory.

The second answer is dynamical: once we identify $\partial L/\partial \dot{q}_i$ as momentum and $\partial L/\partial q_i$ as force, the equation becomes a generalized form of $F = ma$. It says that the correct path, singled out globally by the action, is exactly the path whose local motion satisfies a force law.

The quickest way to understand that statement is now to work through examples.

3.6 Worked Newtonian Examples and Translation Symmetry

We start with the simplest possible system: one particle of mass m moving along a line. The Lagrangian is

$$L = \frac{1}{2}m\dot{x}^2 - U(x). \tag{3.21}$$

Here the generalized coordinate is simply $q = x$. Then

$$\frac{\partial L}{\partial \dot{x}} = m\dot{x}, \qquad \frac{\partial L}{\partial x} = -\frac{dU}{dx}. \tag{3.22}$$

Substituting into the Euler–Lagrange equation gives

$$m\ddot{x} = -\frac{dU}{dx}. \tag{3.23}$$

So the variational formalism reproduces the ordinary Newtonian equation of motion.

At once the lecture inserts a useful warning. If we had chosen $L = T + U$, the force law would have come out with the wrong sign. The Lagrangian is $T - U$, not the conserved energy. Energy conservation will be discussed later, and the conserved quantity there will be $T + U$, not L itself.

For many Cartesian coordinates the same calculation simply repeats coordinate by coordinate. If

$$T = \sum_i \frac{1}{2}m_i\dot{x}_i^2, \qquad L = T - U(x_1, \ldots, x_N), \tag{3.24}$$

then

$$p_i = \frac{\partial L}{\partial \dot{x}_i} = m_i\dot{x}_i, \qquad \frac{dp_i}{dt} = -\frac{\partial U}{\partial x_i}. \tag{3.25}$$

The label i now runs over all Cartesian coordinates of all particles. This is still $F = ma$, but now it is written one component at a time for the entire system.

The next example is where the lecture's deeper theme first becomes visible. Consider two particles on a line with a translation-invariant interaction:

$$L = \frac{1}{2}m_1\dot{x}_1^2 + \frac{1}{2}m_2\dot{x}_2^2 - U(x_1 - x_2). \tag{3.26}$$

Translation invariance means that the potential depends only on the separation of the particles, not on where the pair sits on the line as a whole.

To make the force relation explicit, set

$$d = x_1 - x_2.$$

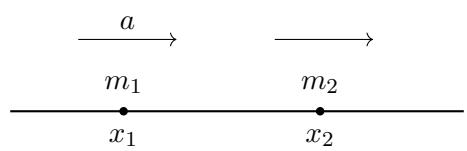


Figure 3.3: A rigid translation moves both particles by the same amount a while leaving the separation $x_1 - x_2$ unchanged.

Then $U = U(d)$, and

$$\frac{\partial U}{\partial x_1} = \frac{dU}{dd}\frac{\partial d}{\partial x_1} = \frac{dU}{dd}, \tag{3.27}$$

$$\frac{\partial U}{\partial x_2} = \frac{dU}{dd}\frac{\partial d}{\partial x_2} = -\frac{dU}{dd}. \tag{3.28}$$

Thus

$$\frac{\partial U}{\partial x_2} = -\frac{\partial U}{\partial x_1}. \tag{3.29}$$

The two Euler–Lagrange equations are

$$\frac{dp_1}{dt} = -\frac{\partial U}{\partial x_1}, \qquad \frac{dp_2}{dt} = -\frac{\partial U}{\partial x_2}. \tag{3.30}$$

Adding them gives

$$\frac{d}{dt}(p_1 + p_2) = 0. \tag{3.31}$$

So the total linear momentum is conserved.

This is the lecture's first fully explicit symmetry-to-conservation argument. Because the action is invariant under a continuous translation, there is a conserved quantity. Here the symmetry is translation invariance, and the conserved quantity is total linear momentum.

The lecture also notes, in passing, that if the potential were made a function of $|x_1 - x_2|$, there would be an inversion symmetry as well. But that is a discrete symmetry, and the present point is the continuous one: symmetries of the action produce conservation laws.

3.7 Two Coordinates, One Broken Symmetry, and Rotational Invariance

The next example is a particle moving near the surface of the Earth. We keep only two coordinates, x horizontally and y vertically. The kinetic term is securely visible on the board, and the transcript gives the potential:

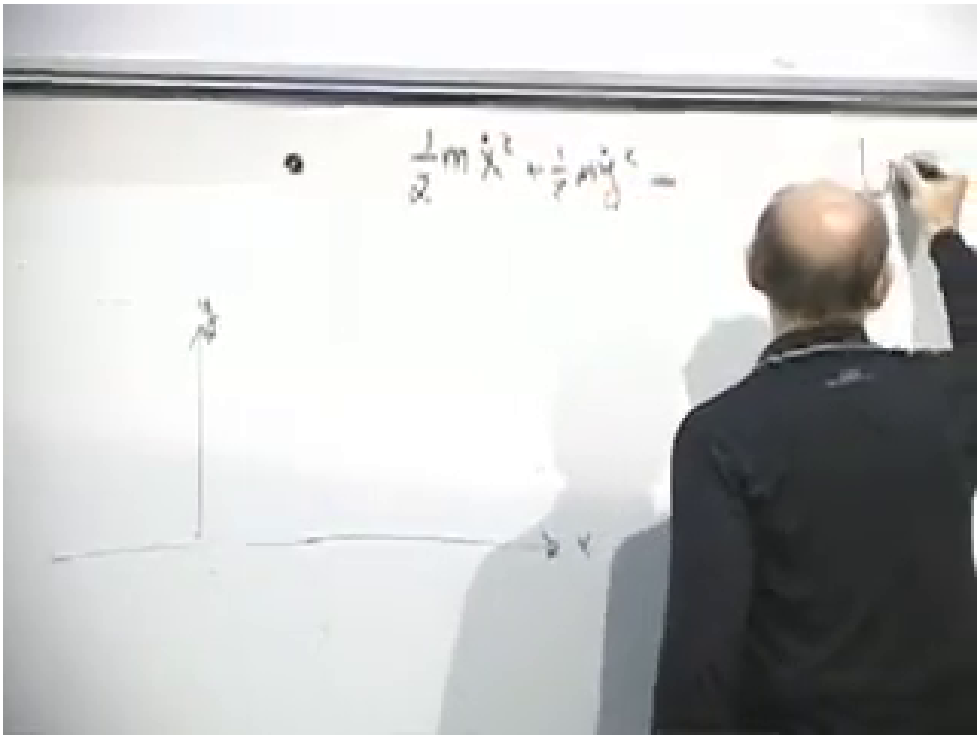
$$T = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}m\dot{y}^2, \qquad U = mgy. \tag{3.32}$$

Thus

$$L = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}m\dot{y}^2 - mgy. \tag{3.33}$$

Because L does not depend explicitly on x, the Euler–Lagrange equation for x gives

$$\frac{d}{dt}(m\dot{x}) = 0. \tag{3.34}$$



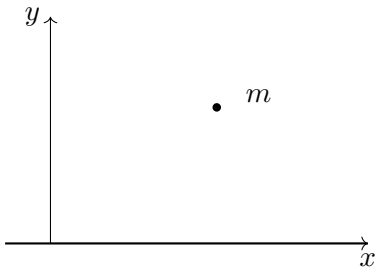


Figure 3.4: The board frame witnesses the two-coordinate kinetic term and the near-Earth geometry. The redraw regularizes only the stable diagrammatic content; the unfinished writing on the far right is intentionally not reconstructed.

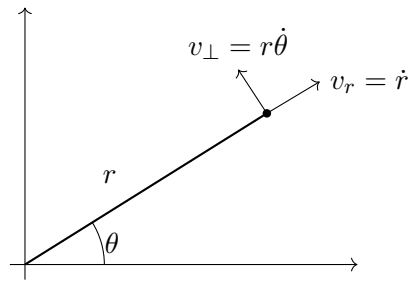


Figure 3.5: Polar-coordinate geometry reconstructed from the lecture's argument. The velocity has a radial component $\dot{r}$ and a perpendicular component $r\dot{\theta}$.

So the horizontal component of momentum is conserved. But for y we have

$$\frac{d}{dt}(m\dot{y}) = -mg. \tag{3.35}$$

There is therefore no conservation of vertical momentum. One symmetry survives; the other does not.

3.7.1 Question & Answer

Why does adding a constant to the potential not restore y-translation symmetry? This is the cleaned mathematical content of the lecture's side discussion. We may always add a constant to U without changing the physics, because the force depends only on derivatives of U. But translating the coordinate y is a different operation. If $U = mgy$, then changing y changes U itself, not merely by a once-and-for-all harmless offset tied to the chosen zero of energy, but by a change that depends on the new value of y. So the Lagrangian is not invariant under y-translations. That is why there is no conserved y-momentum.

This example sets up the lecture's next pivot. The principle of least action is not tied to rectangular coordinates. If we choose another coordinate system, the formulas change clothes, but the action principle itself does not change.

So let us pass to polar coordinates for a particle moving in a plane. The coordinates are r and θ, and the trajectory is now the pair $r(t)$, $\theta(t)$. The lecture derives the kinetic energy geometrically by decomposing the velocity into a radial piece and a perpendicular piece:

$$v_r = \dot{r}, \qquad v_\perp = r\dot{\theta}. \tag{3.36}$$

Therefore

$$T = \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\theta}^2\right). \tag{3.37}$$

Now take a central potential,

$$U = U(r), \tag{3.38}$$

so that

$$L = \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\theta}^2\right) - U(r). \tag{3.39}$$

The transcript becomes noisy in the radial equation, and the lecture's stable emphasis lies elsewhere. The important point is that L does not depend explicitly on θ. Therefore θ is a cyclic coordinate,

and

$$\frac{\partial L}{\partial \dot{\theta}} = mr^2\dot{\theta} \tag{3.40}$$

is conserved:

$$\frac{d}{dt}\left(mr^2\dot{\theta}\right) = 0. \tag{3.41}$$

Hence

$$mr^2\dot{\theta} = \text{constant}. \tag{3.42}$$

This is conservation of angular momentum. It immediately explains the ice-skater effect invoked in the lecture: when r decreases, $\dot{\theta}$ must increase to compensate.

We have thus arrived at the lecture's closing pattern:

translation invariance $\longrightarrow$ linear momentum conservation, rotational invariance $\longrightarrow$ angular momentum co

3.8 Summary

The lecture opens with a grand claim and then earns it step by step. First come two modest mathematical tools: integration by parts and the arbitrary-function lemma. Then comes the variational derivation:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) = \frac{\partial L}{\partial q_i}. \tag{3.43}$$

This equation is the bridge between a global least-action principle and a local differential law.

Only after that bridge is built does the lecture attach physical names to the objects in it. The derivative of the Lagrangian with respect to velocity becomes momentum; the derivative with respect to coordinate becomes generalized force; and the equation begins to read like $F = ma$.

The examples then make the structure concrete. For one particle, the Euler–Lagrange equation reproduces Newton's law. For two interacting particles on a line, translation invariance yields conservation of total momentum. For a central force in polar coordinates, rotational invariance yields conservation of angular momentum. The chapter ends where the lecture ends: not with a single formula in isolation, but with the recognition of a pattern. Symmetry and conservation law belong together, and the principle of least action is the language in which that unity becomes visible.